

# Damien Robert

Deep Learning for Remote Sensing & Environment



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I am a postdoctoral researcher in the [EcoVision lab](#) at [University of Zurich](#), I collaborate with [Jan Dirk Wegner](#) to design deep learning methods for remote sensing and environmental applications. Before joining UZH, I completed my PhD on "[Efficient Learning on Large-Scale 3D Point Clouds](#)" at [IGN](#) and [ENGIE](#), under the supervision of [Loic Landrieu](#) and [Bruno Vallet](#).

## ■ POSITIONS

|           |                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–now  | <b>Postdoctoral Researcher</b><br><a href="#">EcoVision lab, UZH</a> · Zurich, Switzerland<br>Deep learning for remote sensing and environment<br>PI: Jan Dirk Wegner              |
| 2020–2024 | <b>PhD Student</b><br><a href="#">LASTIG, IGN / ENGIE CRIGEN lab</a> · Paris, France<br>Efficient learning on large-scale 3D point clouds<br>Advisors: Loic Landrieu, Bruno Vallet |
| 2017–2020 | <b>R&amp;D Engineer</b><br><a href="#">SIRADEL / ENGIE</a> · Rennes, France<br>Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data                      |
| 2017      | <b>Co-Founder</b><br><a href="#">Inspirama</a> · Rennes, France<br>Website gathering book recommendations from inspiring people                                                    |
| 2015      | <b>R&amp;D Intern</b><br><a href="#">Dassault</a> · Providence, RI, USA<br>Dimensionality reduction and dynamic system modeling                                                    |

## ■ EDUCATION

|           |                                                                                                                                                                                                                 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2011–2015 | <b>MSc, Engineering</b><br><a href="#">École Centrale de Lyon</a> · Lyon, France<br>Mathematics, Computer Science, Mechanics, Signal Processing, Automation                                                     |
| 2022      | <b>International Computer Vision Summer School</b><br><a href="#">International Computer Vision Summer School</a> · Sicily, Italy<br>Computer vision courses by world-renowned experts in academia and industry |
| 2017      | <b>CNRS AI Fall School</b><br><a href="#">CNRS</a> · Lyon, France<br>Multi-disciplinary course for AI students and researchers                                                                                  |

## ■ AWARDS

|      |                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>Outstanding Reviewer</b> · <a href="#">CVPR</a>                                                                 |
| 2025 | <b>PhD Award accessit</b> · <a href="#">AFRIF</a><br>French Association for Pattern Recognition and Interpretation |
| 2024 | <b>Outstanding Reviewer</b> · <a href="#">ECCV</a>                                                                 |
| 2022 | <b>Best Paper Finalist</b> · <a href="#">CVPR</a><br>Top 0.4% of submissions                                       |

2026

**Toward a Foundation Model for Forest Point Clouds**

Y. Yue, S. Puliti, **D. Robert**, A. Topaloğlu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler  
*Under review*

**From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps**

G. Sialelli, R. Young, Y. Jiang, C. Aybar, L. Scheibenreif, **D. Robert**, C. Mosig, M. Czerkawski, A. J. Stewart, J. D. Wegner,  
 A. Pirinen, O. Mogren, K. Schindler  
*Under review*

**Bias Aware Benchmark for Species Distribution Modeling**

E. Arens, N. van Tiel, R. Zbinden, C. Vanalli, **D. Robert**, L. Drees, B. Kellenberger, L. Pellissier, J. D. Wegner, D. Tuia  
[World Biodiversity Forum](#)

**LitePT: Lighter Yet Stronger Point Transformer** Highlight

Y. Yue, **D. Robert**, J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler  
[CVPR](#)

**EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation**

L. Geist, L. Landrieu, **D. Robert**  
[ICRA](#)

**FORMSpot: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series**

M. Schwartz, F. Fogel, N. Besic, **D. Robert**, L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciaia

[Remote Sensing of Environment](#)

2025

**MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation**

K. Karaman, V. S. F. Garnot, **D. Robert**, M. J. Santos, J. D. Wegner  
[PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science](#)

**SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology**

E. Plekhanova, **D. Robert**, J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann  
[CVPR workshop EarthVision](#)

**GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance** Oral

K. Karaman, Y. Jiang, **D. Robert**, V. S. F. Garnot, M. J. Santos, J. D. Wegner  
[ISPRS Geospatial Week](#)

**Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks**

J. Dollinger, **D. Robert**, E. Plekhanova, L. Drees, J. D. Wegner  
[ICLR workshop Tackling Climate Change with Machine Learning](#)

**Lossy Neural Compression for Geospatial Analytics: A Review**

C. Gomes, I. Wittmann, **D. Robert**, J. Jakubik, T. Reichelt, M. Martone, S. Maurogiovanni, R. Vinge, J. Hurst, E. Scheurer,  
 R. Sedona, T. Brunschweiler, S. Kesselheim, M. Batic, P. Stier, J. D. Wegner, G. Cavallaro, E. Pebesma, M. Marszałek, M.  
 A. B. Plomer, K. Adriko, P. Fraccaro, R. Kienzler, R. Briq, S. Benassou, M. Lazzarini, C. M. Albrecht  
[IEEE Geoscience and Remote Sensing Magazine](#)

2024

**Efficient Learning on Large-Scale 3D Point Clouds** AFRIF PhD award

**D. Robert**  
[PhD Thesis, Université Gustave Eiffel](#)

Jury: Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loic Landrieu, Bruno Vallet

**Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering** Oral

**D. Robert**, H. Raguet, L. Landrieu  
[3DV](#) · top 5.3% of submissions

2023

**Efficient 3D Semantic Segmentation with Superpoint Transformer**

**D. Robert**, H. Raguet, L. Landrieu  
[ICCV](#) · top 26.8% of submissions

2022

**Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation** Best paper finalist Oral

**D. Robert**, B. Vallet, L. Landrieu  
[CVPR](#) · top 0.4% of submissions

## ■ COLLABORATION & SUPERVISION

|                 |                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Collaborators   | <b>Konrad Schindler</b> <a href="#">ETH</a> · <b>Christian Rupprecht</b> <a href="#">Oxford</a> · <b>Jan Dirk Wegner</b> <a href="#">UZH</a> · <b>Loic Landrieu</b> <a href="#">ENPC</a> · <b>Vivien Sainte Fare Garnot</b> <a href="#">UZH</a> · <b>Lukas Drees</b> <a href="#">UZH</a> · <b>Martin Schwartz</b> <a href="#">LSCE</a> · <b>Elena Plekhanova</b> <a href="#">WSL</a> |
| PhD Students    | <b>Emilia Arens</b> <a href="#">UZH</a> with Jan Dirk Wegner · <b>Johannes Dollinger</b> <a href="#">UZH</a> with Jan Dirk Wegner · <b>Kaan Karaman</b> <a href="#">UZH</a> with Jan Dirk Wegner · <b>Louis Geist</b> <a href="#">ENPC</a> with Loic Landrieu · <b>Yuanwen Yue</b> <a href="#">ETH</a> with Konrad Schindler and Christian Rupprecht                                 |
| Master Students | <b>Valerio Schelbert</b> <a href="#">WSL</a> · <b>Jackson Sunny</b> <a href="#">École Polytechnique</a> with SAMP                                                                                                                                                                                                                                                                    |
| Advising        | <b>UpCircle AI</b> <a href="#">Switzerland</a> · <b>École des Sciences Criminelles de Lausanne</b> <a href="#">Switzerland</a> · <b>SAMP AI</b> <a href="#">France</a> · <b>GEODIGITAL</b> <a href="#">US</a>                                                                                                                                                                        |

## ■ REVIEWING, CHAIRING & ORGANISING

 Reviewer  Area chair  Organising committee

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2027 |  CVPR workshop EarthVision                                                                                                                                                                                                                                                                                                                                           |
| 2026 |  ECCV ·  CVPR                                                                                                                                                                                                                                                                       |
| 2025 |  CVPR ·  CVPR workshop EarthVision                                                                                                                                                                                                                                                  |
| 2024 |  ECCV ·  ISPRS Journal of Photogrammetry and Remote Sensing ·  ICLR workshop ML4RS ·  CVPR workshop EarthVision |
| 2023 |  CVPR ·  CVPR workshop EarthVision                                                                                                                                                                                                                                                  |
| 2022 |  ISPRS Journal of Photogrammetry and Remote Sensing ·  CVPR workshop EarthVision                                                                                                                                                                                                  |

## ■ TEACHING

|      |                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>Transformer</b> · <a href="#">UZH</a> · Lecture, Master 2                                                         |
| 2025 | <b>AI4Good group projects</b> · <a href="#">UZH</a> · Labs, Master 2                                                 |
| 2025 | <b>Transformer</b> · <a href="#">UZH</a> · Lecture, Master 2                                                         |
| 2024 | <b>Transformer, NeRFs, and Diffusion</b> · <a href="#">UZH</a> · Lecture and labs, Master 2                          |
| 2023 | <b>Deep Learning for Remote Sensing</b> · <a href="#">ENSG-IGN</a> · Lecture, Master 2                               |
| 2022 | <b>3D Deep Learning for Remote Sensing</b> · <a href="#">ISPRS'22</a> · Tutorial, Researchers                        |
| 2022 | <b>3D Deep Learning, Torch-Points3D &amp; DeepViewAgg</b> · <a href="#">ENGIE CRIGEN lab</a> · Tutorial, Researchers |
| 2022 | <b>Deep Learning for Remote Sensing</b> · <a href="#">ENSG-IGN</a> · Lecture, Master 2                               |
| 2020 | <b>Deep Learning for Computer Vision</b> · <a href="#">Ecole Polytechnique</a> · Lecture, Master 1                   |

## ■ OPEN-SOURCE SOFTWARE

|                                                                                                |                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ 406  45   | <b>LitePT</b> ·  <a href="#">prs-eth/LitePT</a>                                        |
| ★ 15  0     | <b>torch-graph-components</b> ·  <a href="#">drprojects/torch-graph-components</a>     |
| ★ 1055  136 | <b>Superpoint Transformer</b> ·  <a href="#">drprojects/superpoint_transformer</a>     |
| ★ 53  3     | <b>NoRA</b> ·  <a href="#">drprojects/nora</a>                                         |
| ★ 74  8     | <b>Point Geometric Features</b> ·  <a href="#">drprojects/point_geometric_features</a> |
| ★ 237  24   | <b>DeepViewAgg</b> ·  <a href="#">drprojects/DeepViewAgg</a>                           |

## ■ SELECTED TALKS

|                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026                                                                                                                   | <b>ESA-NASA workshop on AI Foundation Model for EO</b> · tutorial · Building Earth Observation Embedding Workflows with TerraTorch · Virtual<br><b>ESA Φ-lab</b> · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy                                                                                                                                                                                                                                                                                                                                                   |
| 2025                                                                                                                   | <b>MPI of Animal Behavior</b> · Research at EcoVision lab · Zurich, Switzerland<br><b>Global Forest Watch</b> · Geospatial foundation models for forest disturbance monitoring · Virtual<br><b>NatureFinance</b> · Research at EcoVision lab · Virtual<br><b>NVIDIA Strategic Research Engagement</b> · Superpoint Transformer and EcoVision research · Zurich, Switzerland<br><b>CNES COMET TSI</b> · Remote Sensing for Species Distribution Modeling · Virtual                                                                                                                                     |
| 2024                                                                                                                   | <b>3D Data Academy</b> · interview · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube<br><b>Google DeepMind</b> · Research at EcoVision lab · Zurich, Switzerland<br><b>3DV'24</b> · oral · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland                                                                                                                                                                                                                                                                                          |
| 2023                                                                                                                   | <b>Ecole des Ponts, IMAGINE lab</b> · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France<br><b>National Land Survey of Finland (NLS)</b> · IGN's research on Large-Scale 2D and 3D Learning · Paris, France<br><b>ETH Zürich, Computer Vision and Geometry lab</b> · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland<br><b>ETH Zürich, Photogrammetry and Remote Sensing lab</b> · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland<br><b>Valeo.ai</b> · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France |
| 2022                                                                                                                   | <b>CV News</b> · interview · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US<br><b>CVPR'22</b> · oral · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US                                                                                                                                                                                                                                                                                                                                                     |
| 45 talks given since 2021 — full list at <a href="https://drprojects.github.io/talks">drprojects.github.io/talks</a> . |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## ■ SKILLS & LANGUAGES

|           |                                                                                                                                                         |                                                                                             |                                                                                                       |                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Research  | 3D & geometry · Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning                      |                                                                                             |                                                                                                       |                                                                                                         |
|           | Efficient & scalable learning · Efficient deep learning · Scalable model design · EO data representation and compression                                |                                                                                             |                                                                                                       |                                                                                                         |
|           | Learning paradigms · Multimodal learning · Self-supervised learning · Representation learning                                                           |                                                                                             |                                                                                                       |                                                                                                         |
|           | Environment & Earth observation · Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology |                                                                                             |                                                                                                       |                                                                                                         |
| Tools     |  Python                                                              |  PyTorch |  PyTorch Lightning |  PyTorch Geometric |
|           |  C++                                                                 |  Hydra   |  Weights & Biases  |  Git               |
|           |  Linux                                                               | SLURM                                                                                       |  scikit-learn      |  Plotly            |
|           |  Blender                                                             | LaTeX                                                                                       |                                                                                                       |                                                                                                         |
| Languages | French                                                               | Native                                                                                      | English           | Fluent                                                                                                  |
|           | Spanish                                                              | Intermediate                                                                                | German            | Beginner                                                                                                |
| Abroad    | Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015                                               |                                                                                             |                                                                                                       |                                                                                                         |
| Interests | Nature & forests · Boxing · Mountaineering · Skateboarding · Guitar · Painting · Reading · Board games                                                  |                                                                                             |                                                                                                       |                                                                                                         |